- There are 7 Questions for 35 marks.
- Write your answers neatly, legibly, and logically. Give supporting arguments for your answers. Keep your answers thorough and concise. Your arguments must follow from the basics discussed in class. You may use any result from class directly (without re-deriving).
- Good luck !

1. [5] For each of the functions below, find the derivative
   (a) $f : \mathbb{R}^3 \mapsto \mathbb{R}$ given by $f(x,y,z) = x^2y + x\sqrt{1+z}$
   (b) $f : \mathbb{R}^2 \mapsto \mathbb{R}^2$ given by $f(x,y) = (x+y,\ xy)$
   (c) $f : \mathbb{R}^n \mapsto \mathbb{R}$ given by $f(\underline{x}) = \underline{x}^\top A \underline{x}$
   (d) $f : \mathbb{R}^{n\times n} \mapsto \mathbb{R}$ given by $f(A) = \text{Trace}(A^2)$
   (e) $f : \mathbb{R}^{n\times n} \mapsto \mathbb{R}$ given by $f(X) = \text{Trace}(X^\top AX)$

2. [6] (a) In which direction does the function in Q1 part(a) change the fastest at the origin ?
   (b) What is the Hessian of the function in Q1(c) ?
   (c) Consider the function $f : \mathbb{R}^{n\times n} \mapsto \mathbb{R}$ given by $f(X) = \text{Trace}(X^\top AX) - \text{Trace}(B^\top X)$ for given matrices A, B. You are given that this function has a unique minima. Give a computable expression for the minima $X^\star$ in terms of the matrices A and B. Given the entries of A, B, how many arithmetic operations (in $O()$ notation) are required to compute $X^\star$?

3. [5] Let $p^1 = (p_1^1, p_2^1),\ p^2 = (p_1^2, p_2^2),\ \ldots,\ p^n = (p_1^n, p_2^n)$ be n points in $\mathbb{R}^2$. We want to find the point that minimizes the sum of squared distances to all points p^i. Hence, for any point $x = (x_1, x_2)$ we define the following function:
$$J(x) = \sum_{i=1}^{n} \|x - p^i\|^2.$$
   (a) Calculate $\frac{\partial J}{\partial x_1}(x)$ and $\frac{\partial J}{\partial x_2}(x)$, find the points $x^\star$ where the gradient $\nabla J(x^\star)$ is zero. How many such points are there ?
   (b) Identify these points as maxima, minima or saddle points of J.
   (c) Explain why J has a unique global minima, and find this minima.

4. [6] (a) Evaluate
$$\iint_D \frac{1}{(x^2+y^2)^{n/2}}\, dA,$$
   where n is an integer and D is the region bounded by the circles with center the origin and radii r and R, $0 < r < R$.
   (b) For what values of n does the integral in part (a) have a limit as $r \to 0^+$?

5. [4] Consider a disc in $\mathbb{R}^2$ with center at $(1,0)$ and radius 1. What is the average distance of all the points in the disc to the origin ?

6. [6] (a) Show that
$$\frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \ldots = \ln 2$$
   (b) Find the radius of convergence of the series $\sum_{n=1}^{\infty} n(x+2)^n/3^{n+1}$

7. [3] Evaluate the integral
$$\int_0^1 \int_0^1 e^{\max\{x^2,y^2\}}\, dxdy.$$